

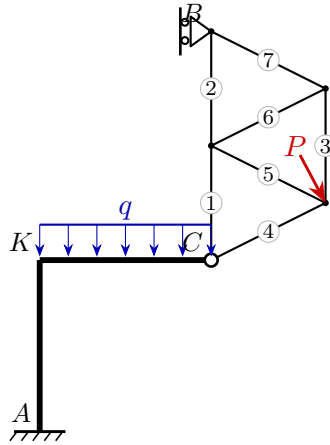
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

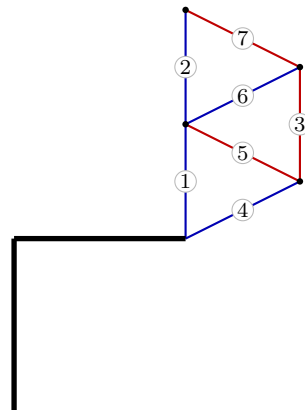
$$A: R_x = 1.5, R_y = 24, M = 54 \quad B: R_x = -9.5, R_y = 0$$

$$C: H_x = -1.5, H_y = -15$$

3. Truss member forces

Member	Force S [kN]	Type
1	-14.25	compression
2	-4.75	compression
3	9.5	tension
4	-1.68	compression
5	10.62	tension
6	-10.62	compression
7	10.62	tension

red: tension (+) blue: compression (–)



4. Section-force diagrams of the beams

